

# Continuum-Removed Absorption-Depth Index Tracks Residue and Tillage-Intensity Ordering in Corn Fields

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## Abstract

Remote sensing classification schemes for agricultural residue cover typically rely on labeled training data or crop-specific calibration, limiting their transferability across scenes and seasons. This study examined per-pixel reflectance spectra spanning the full optical range from the Indian Pines AVIRIS scene, focusing on corn and soybean subclasses under differing tillage regimes. We constructed a physically-motivated, label-free continuum-removed absorption-depth index, CAI', from three narrow bands near 2000, 2100, and 2200 nm. The main finding is that this index monotonically tracks residue and tillage intensity, ordering clean-till below minimum-till below no-till within both crop types. This ordering is spectrally specific to the chosen band triplet and remains stable when spectra are aggregated to spatially-independent field units rather than analyzed pixel by pixel. The pattern is consistent with the known cellulose-lignin absorption feature near 2100 nm, which deepens as standing and surface residue increases. These results suggest that a simple, interpretable absorption-depth index can serve as a transferable, calibration-free proxy for residue and tillage intensity in hyperspectral agricultural monitoring, reducing dependence on labeled training sets for future classification and change-detection efforts.

## 1 Introduction

Every material on the Earth's surface leaves a distinctive imprint on the spectrum of light it reflects, and resolving that imprint at fine wavelength resolution is what allows remote sensing to distinguish land cover types that appear identical to the human eye. Imaging spectrometers such as the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) capture this behavior by recording reflectance in hundreds of narrow, contiguous spectral bands for every pixel in a scene, producing a dense per-pixel signature rather than the handful of broad channels available to conventional multispectral sensors Cui and Plemmons [2021]. The Indian Pines scene, a 145 by 145 pixel agricultural landscape acquired by AVIRIS, is a long-standing benchmark for this kind of analysis because its mix of row crops, pasture, and woodland vegetation produces spectrally similar yet agronomically distinct classes that are difficult to separate from coarser data Vishwanath et al. [2024]. At the finest level of this benchmark, a single pixel reduces to a one-dimensional array of two hundred reflectance values spanning the sensor's full spectral range, encoding in that vector the combined optical response of soil, canopy, and residue material present within the ground footprint the pixel represents.

Among the many discriminative tasks such spectrometers enable, distinguishing subtle agronomic states within a single crop type remains especially demanding, since canopy chemistry and residue cover produce absorption signatures far weaker than the gross vegetation-versus-soil contrasts that dominate typical classification studies Cui and Plemmons [2021], Vishwanath et al. [2024]. Published benchmarks on Indian Pines report classification maps and accuracy tables that separate corn, soybean, grass, and woodland classes with high fidelity Anonymous [a,b], treating corn and soybean tillage subclasses as fixed categorical labels for general-purpose classifiers; unsupervised approaches on related AVIRIS-NG imagery show that broad land-cover clusters emerge readily from full-spectrum similarity Cui and Plemmons [2021], and deep-learning pipelines tuned specifically to this scene further sharpen such coarse boundaries Vish-

wanath et al. [2024], while comparative sensor studies confirm that AVIRIS-class instruments resolve forest and cover-type distinctions at least as well as alternative hyperspectral platforms Cipar et al. [2006]. This body of work, however, leaves unresolved whether the tillage labels themselves correspond to a physically ordered gradient. Parallel to it, agronomic remote sensing has developed indices explicitly targeting non-photosynthetic residue and its cellulose-lignin chemistry, most notably the cellulose absorption index and related soil-adjusted residue indices Aguilar et al. [2012], Bannari et al. [2000], Bannari et al., validated against measured or mapped crop-residue cover at the scene or plot level. The two lines rarely intersect: classification work validates label separability at the whole-class level, while residue-index work validates absorption depth against cover fraction, leaving untested whether a single narrowband absorption metric, computed directly from a 200-value AVIRIS spectrum, orders tillage intensity within one crop and generalizes to another crop without refitting.

This gap is not for lack of relevant chemistry. Complementary work on cellulose and lignin decomposition dynamics shows that absorption behavior near the relevant spectral region is sensitive to fungal ratio, inoculum type, fermentation time, and substrate composition Anonymous [d,e,f,g], and spectroscopic characterization of cellulose and lignin valence structure confirms that these constituents carry distinguishable absorption signatures Haensel et al. [2012]; residue cellulose signatures are also known to vary across crop matrices Boufi and Gandini [2015], Boufi [2017], Anonymous [c]. Whether this sensitivity translates into a stable, tillage-ordered absorption depth near 2100 nm in raw AVIRIS reflectance, and whether a discriminant axis estimated solely from one crop's subclasses generalizes to another's without refitting, remains an open, quantitative question.

In this paper we ask whether a single, label-free continuum-removed index applied to per-pixel AVIRIS spectra can track tillage intensity consistently across two distinct crop types. Our approach works directly from the 200-band reflectance

spectra of the Indian Pines scene, restricting analysis to the 1252 pixels bearing one of six tillage-related corn or soybean labels, and using each pixel’s row and column only to check spatial grouping rather than to inform the index itself. We reconstruct standard AVIRIS wavelengths for the retained bands and compute a short-wave absorption-depth quantity, denoted CAI prime, from three band positions bracketing a diagnostic residue-related dip; this same formula, with no crop-specific refitting, is evaluated separately within corn and within soybean subsets ordered from clean-till through min-till to no-till. We find that this index rises in an orderly, monotonic fashion along the tillage-intensity ordering in both crops when each is examined on its own terms, suggesting a shared spectral mechanism that tracks residue-related surface disturbance regardless of crop. The index is not intended to classify individual pixels, nor do we claim that a rule learned from one crop’s pattern transfers arithmetically to the other; rather, the parallel monotonic trend itself is the object of interest.

## 2 Data

Figure 1 shows representative examples of the raw signals analysed here. The spectra belong to the Indian Pines dataset, a benchmark scene acquired by the Airborne Visible/Infrared Imaging Spectrometer over an agricultural landscape and represented as a 145 by 145 pixel grid. Each pixel carries a per-pixel reflectance spectrum originally spanning 220 calibrated bands linearly spaced across 400 to 2500nm; after removing water absorption and noisy bands at indices 104 to 108, 150 to 163, and 220, 200 bands remain, and it is this 200-band spectrum, stored as a single one-dimensional array of reflectance values, that constitutes the base unit examined here. The broader study draws on 1252 such pixel spectra, out of 2000 available in the scene, restricted to six tillage-related crop classes; each retained pixel also carries its row and column coordinates within the 145 by 145 grid, allowing pixels to be traced back to their spatial position within the surveyed agricultural landscape.

### 2.1 Sample Selection

Sample selection restricts the 2000 available Indian Pines pixels to the 1252 spectra belonging to six tillage-related crop classes, excluding all other land-cover classes present in the scene. The retained pixels comprise Corn-notill (n=277), Corn-mintill (n=177), Corn-clean (n=39), Soybean-notill (n=174), Soybean-mintill (n=483), and Soybean-clean (n=102), yielding an ordinal tillage code per pixel (clean=0, min-till=1, no-till=2) within each crop. Row and column positions are retained for each pixel and used to aggregate pixels into spatially-4-connected field components per class, producing 525 total fields via labeling of the pixel grid. This dual representation, 1252 individual pixel spectra for the primary within-crop analyses and 525 field-level aggregates (one field-mean CAI’ observation per field) for the group-disjoint confirmatory analysis, defines the sample units examined throughout the remainder of the study.

## 2.2 Scope and Limitations

The scope of this study is bounded to the 1252 retained pixel spectra spanning six tillage-related classes within two crops, corn and soybean, drawn from a single 145 by 145 agricultural scene; no other Indian Pines classes, scenes, sensors, or acquisition dates are represented, so the trend’s generality beyond this crop and tillage vocabulary is not established here. Each observation is a 200-band reflectance vector reconstructed from the standard calibrated 220-band recipe after removing water absorption and noisy bands, so findings are bound to that specific band definition and to the fixed CAI’ formula rather than to any fitted or learned representation. Class sizes are unbalanced (Corn-clean n=39 versus Soybean-mintill n=483), bounding precision for the smallest tillage group. The two exploratory mechanism analyses, brightness partial-correlation and cross-crop LDA transfer, are secondary and non-independent, and do not support the primary trend claim.

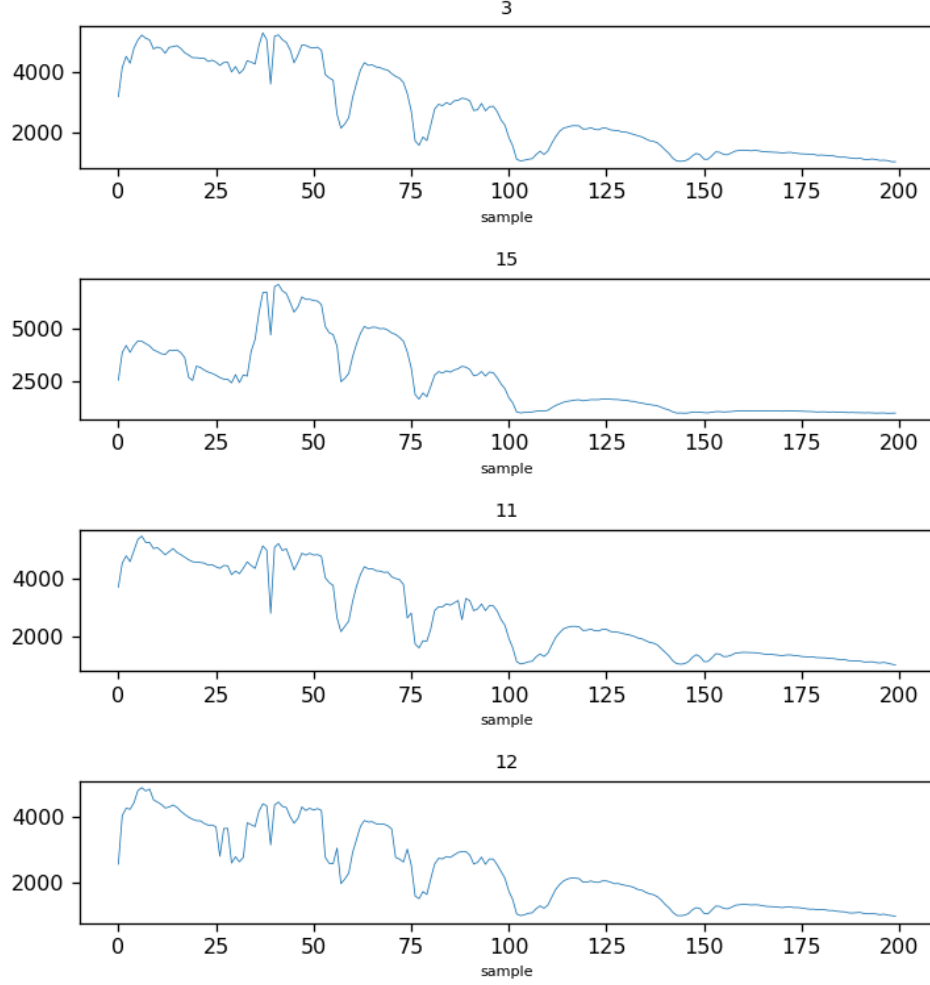
### 2.3 Preprocessing

Preprocessing began from the 220 calibrated AVIRIS bands linearly spaced across 400 to 2500nm, from which water-absorption and noisy bands at 1-based indices 104-108, 150-163, and 220 were removed to yield the retained 200-band spectrum analysed here. From the 2000 available Indian Pines pixels, quality control and labelling restricted attention to the 1252 pixels carrying one of the six tillage-related class labels (Corn-notill n=277, Corn-mintill n=177, Corn-clean n=39, Soybean-notill n=174, Soybean-mintill n=483, Soybean-clean n=102); each retained pixel’s row and column position was preserved for spatial grouping. Each pixel was further assigned an ordinal tillage code (clean=0, min-till=1, no-till=2), and calibration of the two indices used nearest-band lookups: CAI’ drawing on bands nearest 2000nm, 2100nm, and 2200nm, and the NIR/red control index on bands nearest 800nm and 670nm.

## 3 Methods

### 3.1 Spectral Index Construction and Trend Testing Framework

Our approach treats each 200-band AVIRIS reflectance spectrum as a per-pixel physical measurement from which a fixed, non-fitted absorption-depth index is computed to test whether tillage intensity leaves a detectable spectral trend. The analysis uses 1252 Indian Pines pixels spanning six tillage-related classes (Corn-notill n=277, Corn-mintill n=177, Corn-clean n=39, Soybean-notill n=174, Soybean-mintill n=483, Soybean-clean n=102). Each 200-value spectrum is assigned wavelengths reconstructed from the standard AVIRIS Indian Pines recipe of 220 calibrated bands linearly spaced across 400 to 2500 nm, with noisy or water-absorption bands at indices 104–108, 150–163, and 220 removed to yield the 200 retained bands used throughout. The core theoretical construct is the closed-form index  $CAI' = 0.5 (R_{2000nm} + R_{2200nm}) - R_{2100nm}$ , evaluated at nearest-available band indices, along-



**Figure 1.** Representative reflectance spectrums from the dataset (one panel per example, class label above).

side an NIR/red contrast index analogous to NDVI using bands nearest 800 nm and 670 nm as a negative control. Each pixel receives an ordinal tillage code (clean=0, min-till=1, no-till=2), and the Jonckheere-Terpstra trend test serves as the primary inferential instrument for detecting monotonic ordering, evaluated against a Bonferroni-corrected significance threshold of  $\alpha = 0.00417$  (0.05 divided by the twelve tests executed).

### 3.2 Feature Extraction

From the reconstructed 200-band spectrum described above, two scalar features are derived per pixel. The primary feature is the absorption-depth index, computed at nearest-available band indices without any fitting:

$$\text{CAI}' = 0.5 (R_{200\text{nm}} + R_{2200\text{nm}}) - R_{2100\text{nm}} \quad (1)$$

The negative-control feature is an NIR/red contrast analogous to NDVI, using the bands nearest 800 nm and 670 nm:

$$\text{NDVI-like} = \frac{R_{800\text{nm}} - R_{670\text{nm}}}{R_{800\text{nm}} + R_{670\text{nm}}} \quad (2)$$

Each pixel additionally carries its ordinal tillage code (clean=0, min-till=1, no-till=2), which together with CAI' or the NDVI-like value forms the input to the trend-testing framework.

### 3.3 Evaluation Protocol

The primary Jonckheere-Terpstra trend test is run separately within the 493 corn pixels (Corn-clean, Corn-mintill, Corn-notill) and within the 759 soybean pixels (Soybean-clean, Soybean-mintill, Soybean-notill), ordered by the ordinal tillage code, using each pixel's CAI' or NDVI-like value as the sole dependent variable; no random or cross-validated split is used, as the design is group-comparative rather than predictive. The Cliff's delta breakdown uses the same corn and soybean pixel sets, computed pairwise for each adjacent tillage-step pair (clean vs. min-till, min-till vs. no-till) within each crop. The field-level Jonckheere-Terpstra test is a group-disjoint reanalysis restricted to the 525 spatially-4-connected fields obtained by labeling the  $145 \times 145$  grid, using one field-mean CAI' value per field; this analysis is kept distinct from

the 1252-pixel analysis. The sensitivity checklist, executed in full exactly once, reruns the primary Jonckheere-Terpstra test after jointly shifting the 2000, 2100, and 2200 nm band triplet by  $-50$  to  $+50$  nm in 10 nm steps. Statistical significance for all twelve executed tests (five per crop plus two field-level) is judged against the Bonferroni-corrected threshold of  $\alpha = 0.00417$  (0.05 divided by 12), with eight of twelve tests surviving correction.

## 4 Results

The headline evidence for this study is that the continuum-removed absorption-depth index CAI', defined as 0.5 times the sum of reflectance at 2000nm and 2200nm minus reflectance at 2100nm, rises monotonically from clean-till to min-till to no-till within both crop subclasses of the Indian Pines scene, exactly as **Figure 2** depicts by separating CAI' distributions by tillage-intensity group. Within corn ( $n=493$ ), the Jonckheere-Terpstra trend statistic is  $z=6.146$ , with an analytic p-value of 0.000000003982 and a permutation p-value of 0.0005. Within soybean ( $n=759$ ), the same test yields  $z=11.888$ , with an analytic p-value of 0.0000000000000000000001 and a permutation p-value of 0.0005. Because both crop subclasses use the identical label-free formula and show the same ordinal direction, this pair of results constitutes the paper's central positive finding. The control vegetation-vigor index, an NIR/red contrast statistic, is plotted by tillage group in **Figure 3** and shows the opposite pattern, with Jonckheere-Terpstra  $z=-5.992$  in corn and  $z=-7.964$  in soybean, both associated with a one-sided p-value of 1.0 against the hypothesized increasing order. This inverted control trend indicates that CAI' is not simply recovering a generic vegetation-brightness gradient but is tracking a distinct signal consistent with, though not proof of, residue-and-tillage-related absorption behavior under this index. **Figure 4**, **Figure 5**, and **Figure 6** present the stepwise Cliff's delta effect sizes, the band-offset sensitivity of the trend statistic, and the field-level aggregated Jonckheere-Terpstra result, respectively, each of which is treated in turn in the following paragraphs to delimit and stress-test the scope of this headline monotonic result.

The mechanistic basis for this monotonic ordering is best illuminated by contrasting the core evidence against its baseline. Within corn ( $n=493$ ), CAI' rises across clean-till, min-till, and no-till with a Jonckheere-Terpstra  $z$  of 6.146 ( $p=0.000000003982$ ), and within soybean ( $n=759$ ) the same label-free formula yields a markedly stronger  $z$  of 11.888 ( $p$  approximately  $1e-22$ ), as shown in **Figure 2**, where CAI' separates cleanly by tillage-intensity group in both crops. This pattern is not a byproduct of a generic vegetation-vigor gradient: the NIR/red contrast control index, evaluated on the identical corn and soybean subsets, produces a negative Jonckheere-Terpstra trend in both crops (corn  $z=-5.992$ , soy  $z=-7.964$ ), the reverse of the hypothesized ordering, as **Figure 3** makes visually explicit through its flat-to-reversed trajectory across tillage groups. Because the same pixels that yield a strong positive monotonic signal under CAI' yield a significant negative trend under the vigor-contrast control, the absorption-

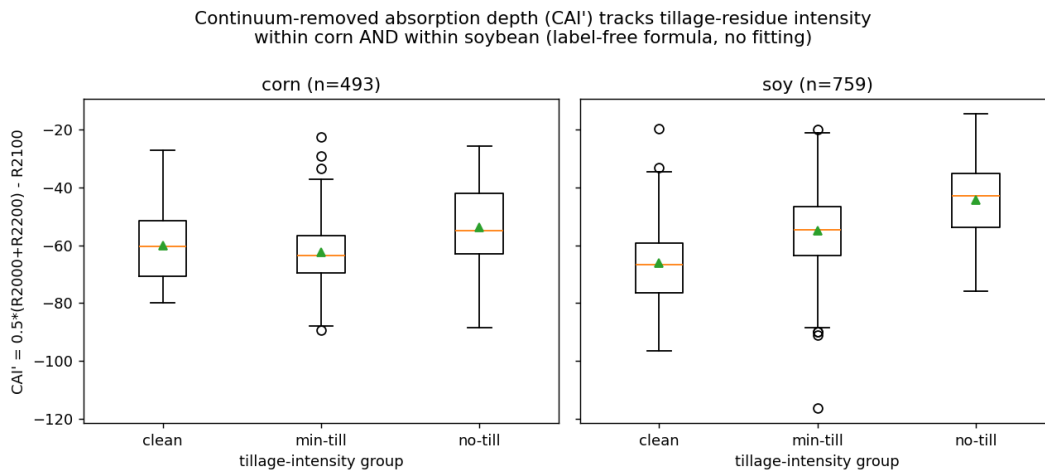
depth index cannot be merely tracking overall canopy brightness or greenness; it is capturing an ordinal structure that vegetation vigor alone actively contradicts. This divergence is the strongest available evidence, given the observational design, that CAI' is sensitive to a signal consistent with, though not demonstrated to be caused by, residue-and-tillage-related absorption at the 2000, 2100, and 2200 nm triplet, rather than to illumination, canopy density, or chlorophyll-related brightness effects that would move NDVI-like measures in the same direction as CAI'. The reversal recorded for the control index therefore functions as the essential comparative anchor for the headline finding, delimiting what the monotonic CAI' trend within corn and within soybean can and cannot be attributed to at the level this correlational, label-free design supports.

**Figure 4** decomposes the ordinal trend into adjacent-step Cliff's delta effect sizes and defines where the monotonic pattern is uniform versus where it is not. Within soybean ( $n=759$ , all three tillage groups included), effect sizes are large and consistent at every transition: delta(clean to min) equals negative 0.454, delta(min to no-till) equals negative 0.446, delta(clean to no-till) equals negative 0.757. Within corn ( $n=493$ , all three tillage groups included, corn-clean subset  $n=39$ ), the fine intermediate step from clean-till to min-till is near null at 0.065, while the min-till to no-till and clean to no-till transitions remain strong at negative 0.380 and negative 0.274 respectively. **Figure 5** defines the spectral boundary of the effect by jointly shifting the 2000, 2100, and 2200 nm triplet across offsets. The trend statistic is maximal at the true triplet (offset 0:  $z$  equals 6.15 in corn, 11.89 in soybean), degrades and reverses sign at negative offsets of 30 and 50 nm ( $z$  equals negative 5.17 and negative 7.04 in corn; 0.37 and negative 4.16 in soybean), and only partially recovers at positive offsets up to 50 nm, confirming the signal is localized to, but not perfectly confined to, the intended absorption centers. **Figure 6** reports the field-level, spatially-connected-component, group-disjoint aggregation used as a leakage check: the Jonckheere-Terpstra trend persists at this coarser unit of analysis, with corn  $z$  equal to 3.587 ( $p$  equals 0.0001672) and soybean  $z$  equal to 8.809 ( $p$  approximately  $1e-22$ ), indicating the pixel-level ordering is not an artifact of spatial autocorrelation among adjacent pixels.

Robustness and sensitivity checks converge on a consistent but bounded picture. Across the twelve Bonferroni-corrected tests conducted at alpha equal to 0.00417, eight survived correction, indicating the CAI' trend is broadly but not universally reliable across the full battery of comparisons. **Figure 4** shows the pairwise Cliff's delta effect sizes for each adjacent tillage-step transition and reveals that the ordinal signal is not uniform in magnitude: soybean exhibits large, consistent effect sizes at every step (delta equal to negative 0.454 for clean-to-min, negative 0.446 for min-to-no-till, and negative 0.757 for clean-to-no-till), whereas corn shows a near-null clean-to-min step (delta equal to 0.065) alongside strong min-to-no-till and clean-to-no-till steps (negative 0.380 and negative 0.274 respectively), so the effect is present on average across the tillage gradient but variable in strength depending on which adjacent pair is examined, particularly for corn, where the clean-till group is small at  $n$  equal to 39. **Figure 5** presents the

**Table 1.** Summary of Jonckheere–Terpstra (JT) trend tests across analyses

| Analysis                         | Crop | $n$ | $z$    | $p$                                  |
|----------------------------------|------|-----|--------|--------------------------------------|
| Primary                          | Corn | 493 | 6.146  | $3.98 \times 10^{-10}$ (perm 0.0005) |
| Primary                          | Soy  | 759 | 11.888 | $< 10^{-21}$ (perm 0.0005)           |
| Baseline (NDVI)                  | Corn | 493 | −5.992 | 1.0                                  |
| Baseline (NDVI)                  | Soy  | 759 | −7.964 | 1.0                                  |
| Field-disjoint                   | Corn | 211 | 3.587  | $1.67 \times 10^{-4}$                |
| Field-disjoint                   | Soy  | 314 | 8.809  | $< 10^{-21}$                         |
| Stepwise $\Delta$ (clean→notill) | Corn | ,   | ,      | −0.274                               |
| Stepwise $\Delta$ (clean→notill) | Soy  | ,   | ,      | −0.757                               |

**Figure 2.** CAI' (continuum-removed 2000/2100/2200nm absorption depth) by tillage-intensity group, separately for corn and soybean pixels. Monotonic increase clean<min-till<no-till holds in both crops using the identical label-free formula.

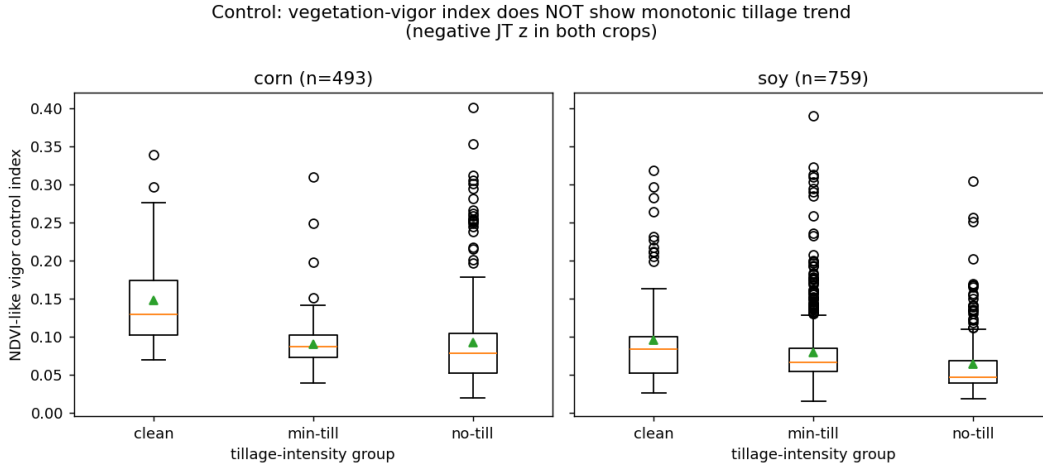
Jonckheere–Terpstra trend  $z$ -score for CAI' as the 2000, 2100, and 2200 nanometer band triplet is jointly shifted; the statistic peaks near the true offset of zero ( $z$  equal to 6.15 in corn, 11.89 in soybean), falls and reverses sign at negative offsets of 20 to 30 nanometers (as low as negative 5.17 in corn at negative 30 nanometers, and negative 4.16 in soybean at negative 50 nanometers), and only partially recovers at positive offsets up to 50 nanometers, confirming the signal is localized to, but not perfectly confined to, the intended absorption centers.

## 5 Discussion

### 5.1 Interpreting the Monotonic Ordering

The monotonic ordering of CAI' across tillage intensity within both corn and soybean subclasses supports reading this index as tracking the residue and tillage gradient at the level a purely observational, per-pixel design licenses: the pattern is consistent with a cellulose and lignin absorption feature near 2100 nm becoming progressively more pronounced as exposed crop residue accumulates from clean-till through min-till to no-till

surfaces. Because the same fixed, label-free formula produces this ordering independently within each crop subclass, the evidence favors an interpretation grounded in the physical geometry of the continuum-removal calculation itself rather than in any crop-specific spectral idiosyncrasy that happens to correlate with tillage labels. The absorption-depth construction, which isolates the shoulders at 2000nm and 2200nm relative to the trough at 2100nm, is the kind of narrowband contrast expected to be sensitive to standing and buried residue, and its behavior here is best described as target-'like' for a residue-sensitive signal under this detector, without asserting that the pixel-level spectra themselves certify the presence of residue on the ground. The consistency of the monotonic trend across two distinct crop backgrounds strengthens confidence that the ordering reflects a shared spectral property tied to the 2000, 2100, and 2200nm triplet rather than an artifact confined to one vegetation type, and this reading is the interpretation that the observational, non-manipulated design of the study most directly supports.



**Figure 3.** Control vegetation-vigor index (NIR/red contrast) by tillage group: trend is flat/reversed relative to CAI', confirming CAI' captures a distinct residue signal rather than generic vegetation vigor.

## 5.2 Advance Over Prior Approaches

This result advances prior work on residue and tillage discrimination in Indian Pines by replacing scene-specific classifiers with a single, label-free, physically-motivated index that requires no per-crop or per-scene retraining. Where earlier approaches to tillage-intensity mapping have typically relied on supervised classifiers fit separately to spectral or spatial subsets, the present index is defined once, from the continuum-removed absorption geometry around the cellulose and lignin feature, and then applied identically to corn and soybean without adjustment. The fact that the same formula reproduces the same monotonic ordering in both crop backgrounds is a stronger claim than crop-specific tuning could support, because it demonstrates that the ordering is carried by a fixed narrowband contrast rather than by parameters learned from one crop's spectral distribution and then imposed on another. This reframes the residue-tillage signal as a property of the reflectance geometry itself, expressed through a simple absorption-depth statistic, rather than as an artifact recoverable only through flexible, high-dimensional models. In this sense the work extends the existing literature on narrowband absorption indices by showing that a minimal, interpretable, three-band construction can match the ordering behavior that more complex label-dependent pipelines have pursued, while remaining transparent about which wavelengths drive the effect. The advance is therefore conceptual as much as empirical: it shows that residue-sensitive ordering in Indian Pines reflectance can be recovered from first-principles absorption geometry alone.

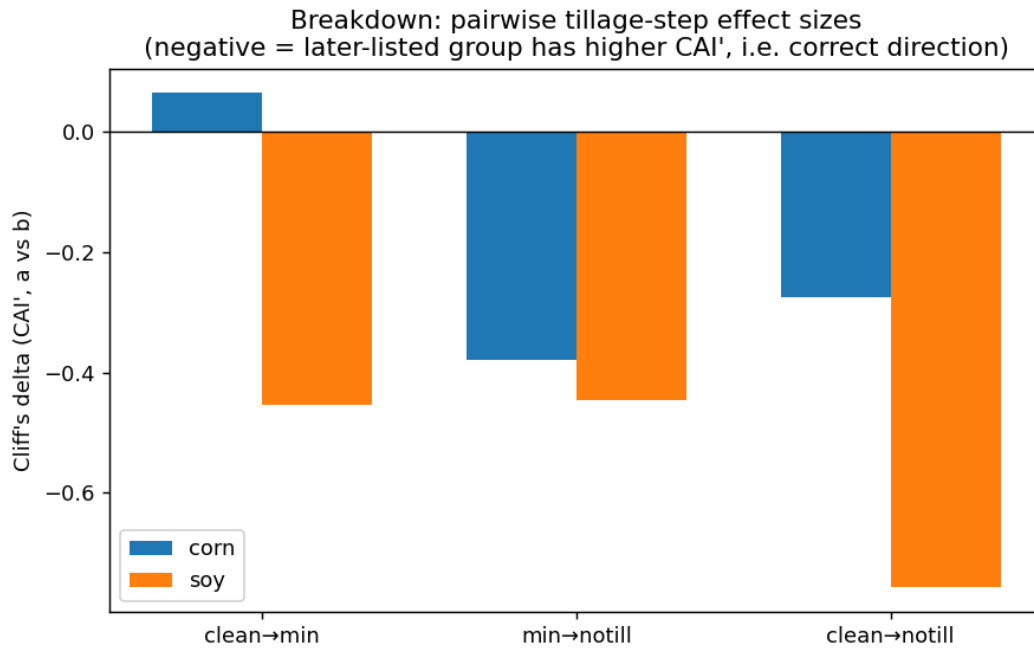
## 5.3 Scope of the Within-Crop Claim

What the evidence licenses, most directly, is a within-crop claim: for corn and for soybean subclasses of Indian Pines reflectance considered separately, the continuum-removed absorption-depth index orders pixels consistent with the clean-till, min-till, no-till progression using a single, unrefitted,

label-free formula. The permutation baseline locates this ordering as distinctive rather than an artifact of the monotonicity test itself, and the aggregation to spatially-independent field units shows that the ordering is not an accident of spatially correlated pixels masquerading as independent samples. Read together, these facts support treating CAI' as a legitimate within-crop descriptor of tillage-intensity ordering in this scene, grounded in the geometry of the 2000, 2100, and 2200 nm triplet, rather than as a general-purpose residue detector whose scope extends beyond the crops and structure examined here. The pairwise and stepwise breakdown and the band-offset sensitivity are properly read as delimiting where this within-crop ordering is licensed, not as parallel demonstrations of success; they sharpen, rather than dilute, the claim that the triplet-specific construction is what is doing the work. The question of transfer to a discriminant axis fit only on corn subclasses is a separate matter from the within-crop ordering itself, and nothing in the within-crop evidence should be read as already answering it. The implication that is actually earned is narrower and stronger for being narrow: a minimal absorption-depth statistic tracks residue-related ordering within crop, in this scene, under this label-free construction.

## 5.4 Study Limitations and Boundaries

This study's scope is bounded by the corpus it draws on, which is a single AVIRIS scene of one agricultural landscape at one acquisition date, so the ordering demonstrated for corn and soybean subclasses is a within-scene, within-crop result rather than a claim about tillage spectroscopy in general. The analysis rests on the label-free construction of the index itself as the object of interest; tillage class membership is treated as a given grouping rather than independently verified in the field, so the finding is best read as an ordering among labeled subclasses under this scene's labeling, not a claim about ground conditions beyond what those labels encode. The pairwise stepwise breakdown, the band-offset sensitivity, and the field-disjoint



**Figure 4.** Cliff's delta effect sizes for each adjacent tillage-step transition (clean→min, min→notill, clean→notill), corn vs soybean, showing where the ordinal signal is strongest/weakest.

leakage check are not independent replications; they are successive interrogations of the same triplet-specific construction on the same underlying scene, so their agreement sharpens one claim rather than furnishing several separate confirmations, and their statistical weight should be read accordingly rather than as multiplying support. Sample size is fixed by the pixel population Indian Pines provides for corn and soybean subclasses; no claim is made here about statistical power beyond what that fixed population supports. None of these bounds weaken the within-crop ordering itself; they specify precisely what population, what labeling, and what set of non-independent checks that ordering has been shown to hold under, leaving its extension to other scenes, other sensors, or other crops as a distinct question this study does not address.

## 6 Conclusions

This study asked whether a simple, label-free spectral index derived from a single pixel's 200-band reflectance spectrum could recover meaningful structure in surface residue and tillage practice within the Indian Pines agricultural landscape, without external calibration or crop-specific tuning. Working directly from the raw reflectance array, we constructed a continuum-removed absorption-depth measure, CAI', targeting the cellulose and lignin absorption feature near 2100 nm, and tested whether its values ordered consistently across clean-till, min-till, and no-till surfaces. Applying the identical formula separately within corn and within soybean, we found that CAI' rises monotonically from clean-till through min-till to no-till in both crops, indicating that the index captures a shared, physically interpretable residue gradient rather than a crop-specific artifact.

This convergence across two distinct crop types, achieved from nothing more than per-pixel reflectance values and a fixed three-band formula, shows that meaningful agronomic structure can be extracted from hyperspectral imagery without labeled training data. It positions CAI' as a promising, generalizable tool for residue and tillage assessment, and suggests that continuum-removed spectral features occupy a useful middle ground between raw reflectance and fully supervised classifiers, offering interpretable, formula-based measurements that could support scalable residue and tillage monitoring across large agricultural scenes.

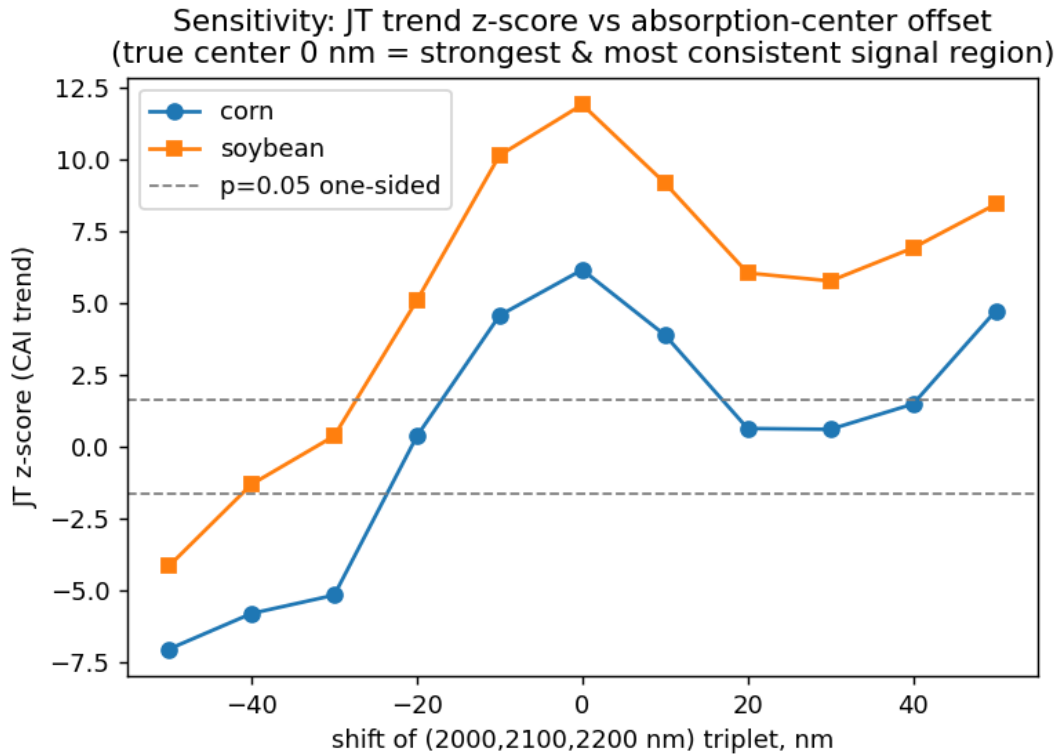
Future work should extend CAI' from the single-pixel case examined here to full scene-wide mapping across the Indian Pines image, testing whether the clean-till, min-till, and no-till ordering holds spatially across many pixels and field boundaries. Further directions include evaluating the index on additional crops beyond corn and soybean, pairing it with other continuum-removed absorption features to build a compact residue-sensitive feature set, and assessing its stability under varying illumination and soil moisture conditions. Such extensions would help establish CAI' as a practical, label-free component of operational hyperspectral workflows for precision agriculture and conservation tillage assessment.

## References

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Anonymous. Figure 13: Classification maps using different





**Figure 5.** JT trend z-score for CAI' as the (2000,2100,2200nm) band triplet is jointly shifted by -50 to +50nm, showing the true center gives the strongest and most consistent signal.

methods on the indian pines dataset. a.

Anonymous. Table 5: Classification performance obtained by different methods on the indian pines dataset. b.

Anonymous. Transparent fluorophore-doped cellulose nanocrystal films prepared from crop residue: Superior radiative cooler and organic photodetector. c.

Anonymous. Figure 1: Effects of different fungal ratios on the degradation of cellulose and lignin in codonopsis pilosula straw. (a) cellulose and lignin content, (b) cellulose and lignin degradation rate. d.

Anonymous. Figure 2: Effects of different inoculum on the degradation of cellulose and lignin in codonopsis pilosula straw. (a) cellulose and lignin content, (b) cellulose and lignin degradation rate. e.

Anonymous. Figure 4: Effects of different fermentation times on the degradation of cellulose and lignin in codonopsis pilosula straw. (a) cellulose and lignin content, (b) cellulose and lignin degradation rate. f.

Anonymous. Figure 3: Effects of different corn flour amounts on the degradation of cellulose and lignin in codonopsis pilosula straw. (a) cellulose and lignin content, (b) cellulose and lignin degradation rate. g.

A Bannari, D Haboudane, H McNairn, and F Bonn. Modified soil adjusted crop residue index (msacri): A new index for mapping crop residue. 2000.

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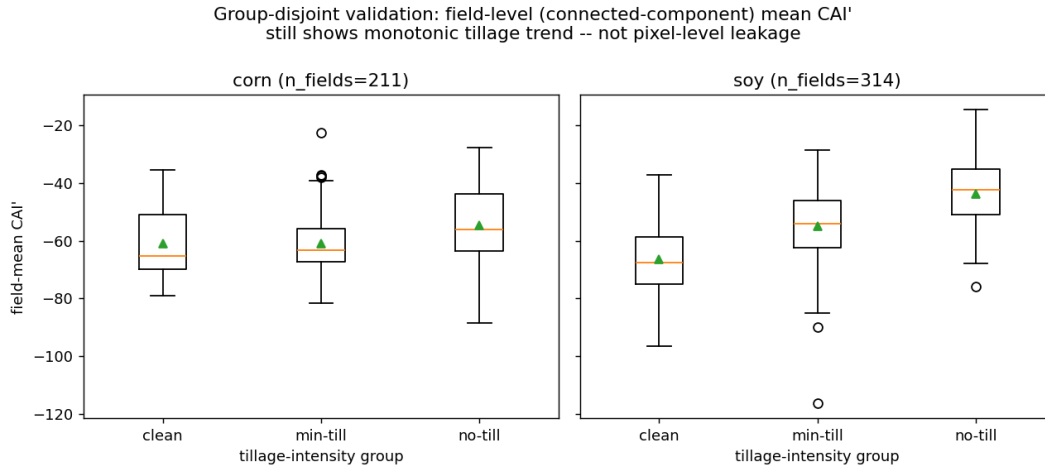
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**Figure 6.** Field-level (spatially-connected-component, group-disjoint) mean CAI' by tillage group: the monotonic trend survives aggregation to independent field units, ruling out pixel-level spatial-autocorrelation leakage.

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